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Agent Souper

A Liquid Agent-System Substrate for Studying Telehomeostasis and the Emergence of Meta-Agents

An open Kolmogorov-Theory simulation platform (Beta)

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Abstract

Agent Souper is an open, local-first simulation platform (currently in Beta) for *liquid agent systems*: collections of N Kolmogorov-Theory (KT) algorithmic agents interacting in a shared 2D world through a *dynamical connectome* $W(t)$, with first-class world objects and niche construction. It is the executable counterpart of the *Agent Soups* formalism[1] and is built to the bcom-handbook L1/L2/L3 agent taxonomy. Every L3 agent is an explicit KT decomposition — persona plus the seven modules (world model, self model, objective function, simulator, comparator, updater, planner) plus a swappable I/O membrane — running a strict *dual-rate* cognition contract: a deterministic fast loop (enumerate → score through the objective function → argmax, every tick, never an LLM) and an optional, per-agent, fire-and-forget slow loop in which a language model acts as a slow objective-function modulator, valence sensor, and narrative world-model maintainer, reaching the fast loop through a single channel and never blocking it. We use the platform to study two questions central to KT: *(i)* the role of *telehomeostasis* — a collective, lineage-level persistence objective compressed into per-agent rules[2] — and *(ii)* the conditions under which a coordinated collective behaves as a coherent higher-order agent (a *meta-agent*). Two pre-registered experiments are reported honestly, including a refutation and a methodological lesson about substrate-dependent artifacts: in 11-v-11 soccer a telehomeostatic "team-scoring" objective beats an otherwise identical selfish objective 7/8 seeds (23–2 aggregate, $\sim 11\times$), but only once the substrate can no longer flatter it; in predator–prey, a sacrificial draw-fire prey objective *reduces* species persistence ($0.57\times$ the selfish baseline, 0/8 seeds) in the low-predation regime — the value of a collective-framed objective is contingent on the environment's binding constraint. We argue that meta-agent emergence is best studied not by building a central orchestrator but by giving each local agent a shared observable and a compressed collective objective, and asking when coordinated agency emerges from purely local reasoning.

Keywords: Kolmogorov Theory, algorithmic agents, agent soups, dynamical connectome, telehomeostasis, meta-agents, dual-rate cognition, objective function, multi-agent simulation, emergence

Classification: ACM CCS: Computing methodologies → Multi-agent systems / MSC 68T42, 68Q30

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1 Introduction

Kolmogorov Theory (KT) models cognition as *algorithmic agency*: an information-processing system that infers and runs compressive world models, evaluates a scalar objective, and plans actions that maximise it[3,4]. The *Agent Soups* programme extends this from a single agent to populations of N agents coupled through a *dynamical connectome* $W(t)$ and embedded in a world they can themselves modify[1]. Two questions sit at the centre of that programme. First, the role of *telehomeostasis* — the generalisation of life as the maintenance of a pattern across computations, scaled from the individual to the lineage or collective[2]. Second, whether and when a coordinated collective is itself an algorithmic agent — a *meta-agent* — in the precise KT sense.

These questions are dynamical and multi-scale; they resist purely analytic treatment. **Agent Souper** is the executable substrate we built to study them empirically. It is an open, local-first simulation platform — currently a working **Beta** — in which KT agents of graded sophistication (the bcom-handbook L1/L2/L3 taxonomy) interact in a shared 2D world with first-class objects, obstacles, and environmental dynamics. It is deliberately not a bespoke demo: every agent–environment interaction is a typed contract, every L3 agent is an explicit KT decomposition, and the cognitive core is deployment-agnostic.

Contributions. (1) An open KT-faithful agent-soup runtime with a strict dual-rate cognition contract and an optional, per-agent, LLM slow loop that modulates — never replaces — the rule-based core. (2) A concrete operationalisation of telehomeostasis as a *compressed per-agent objective plus a shared observable*, with no central orchestrator. (3) Two honestly-reported experiments, including a refutation and a methodological result: the sign of a collective-objective effect can be a substrate artifact, and only survives belief once the substrate can no longer flatter it. (4) A research framing of meta-agent emergence as a question about local reasoning toward a shared goal, not about central control.

2 The KT agent inside Agent Souper

An algorithmic agent in KT is the triad *Model Engine* (ME), *Objective Function* (OF), and *Planning Engine* (PE): a compressive world model, a scalar valuation, and an action selector $a_{t+1} = \arg\max_{a'} \mathbb{E}[\,O(M'(m_t, a'))\,]$[3]. Agent Souper realises this as the bcom-handbook **L3 skeleton**: a `persona` plus seven modules — `world_model`, `self_model`, `objective_function`, `simulator`, `comparator`, `updater`, `planner` — and a swappable `membrane` that is the only deployment-specific surface. Each module ships a Markdown specification beside its code stating its KT role; the specification is the contract, the code one implementation of it. The `persona` is the front door (identity, voice, values); an empty persona is the reliable signature of an L2 agent dressed up as L3. This is not an analogy: the seven modules are seven concrete Python modules under `agents/kt/`, constructed and wired by `agents/algorithmic.py`, with their KT-role specifications in `agent_template/**/*.md`. Figure 1 is the map.

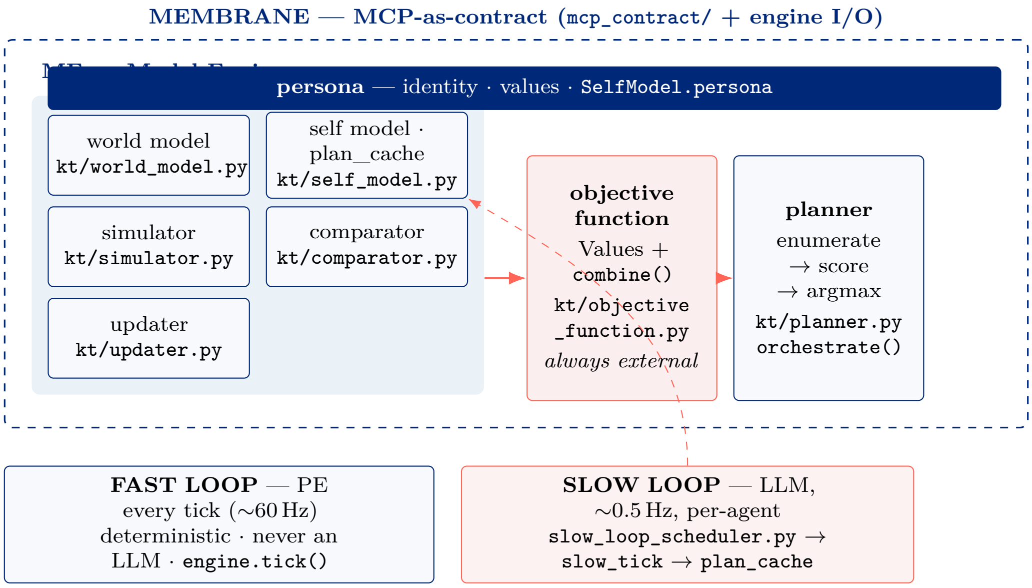

Figure 1: The KT agent model mapped one-to-one onto the Agent Souper software. ME / OF / PE is the seven-module L3 skeleton; every module is a concrete file under `agents/kt/`; the slow loop modulates ME/OF/PE only through the single `plan_cache` channel and the engine never blocks on it. *We are running the agent architecture, not approximating it.*

The taxonomy is graded. An **L1** `Actor` is a pure observation→action policy (no objective function, no LLM); **L2** (a fused neural policy with external reward) is defined but deliberately not yet implemented; **L3** is the full decomposition above. A non-negotiable invariant carries over from the formalism: the *objective function is always external* — never fused into a network. Fusing it is precisely what demotes an agent from L3 to L2. This keeps the agent's values inspectable and editable, which is what makes the experiments below interpretable rather than black-box.

3 Dual-rate cognition: rules fast, LLM slow

The defining engineering and conceptual commitment of Agent Souper is a strict **dual-rate** contract.

The fast loop runs every tick (~60 Hz), is deterministic under a seed, and never calls a language model. The planner enumerates a closed set of candidate actions; the objective function scores each candidate as a priority-weighted, normalised sum of per-Value scores $s_v(a) \in [-1, 1]$,

$$\mathrm{score}(a) \;=\; \frac{\sum_v w_v\, s_v(a)}{\sum_v w_v}, \qquad w_v \;=\; 0.5^{\,(\mathrm{priority}_v - 1)} \cdot r_v, \tag{1}$$

and the planner takes the arg max with a seeded tie-break. Priority 1 dominates priority 2 (weight 1.0 vs 0.5) until it saturates, so *the objective-function stack is the behaviour.* The multiplier r_v defaults to 1.

The slow loop is optional, fire-and-forget, per-agent, and runs at ~0.5 Hz. A scheduler calls the agent's slow tick and returns immediately — *the engine never awaits the LLM.* The slow tick reaches the fast loop through exactly one channel, the agent's `plan_cache`, and only there. Through it the LLM performs three KT-aligned roles, and *no others*:

- **Slow OF re-weighting** — it writes r_v (the `value_reweights`, clamped to $[0, 5]$), the per-Value multipliers in Eq. (1). This *does* change the Objective Function's evaluation: the scalar the planner argmaxes shifts, because the value-aggregation weights change. What it does *not* touch is the per-Value score functions $s_v(a)$ (fixed rules); it cannot add, remove, or redefine a Value, and it cannot name an action. So it reshapes the landscape; the rules still climb it.

- **Slow PE self-tuning** — it writes a small, allowlisted, clamped set of its own planner parameters (`plan_params`: subgoal-pull strength, step-size multiplier) and an optional `subgoal` the planner adds a soft attraction term toward.
- **Valence sensing & narrative ME** — it synthesises an `official_valence`/`arousal` reading (display and logging only — it never feeds action selection) and maintains a rolling natural-language world-model that is re-fed to it as memory.

This LLM–OF coupling is one implementation option

KT requires only that the Objective Function be *external* and *scalar*; it does *not* prescribe whether or how a slow process modulates it. Re-weighting the combine is the design choice *Agent Souper currently makes* — equally KT-valid alternatives include leaving the OF fixed, having the slow loop propose new Values through the closed-loop updater (placement test #4), or having it tune the model/planner instead. The choice is a research variable, not a commitment of the theory.

Three properties make this a clean experimental instrument rather than an uncontrolled confound. *(i) Identity fallback*: with the LLM off (or before its first emission) the cache is identity, $r_v \equiv 1$, no subgoal — behaviour is *exactly* the rules-only baseline, and the persona is completely inert (it is read in one place only, to build the LLM prompt). *(ii) Per-collective gating*: the LLM is enabled per *collective* (predators, prey, or a soccer team) while reasoning stays strictly per-agent — "LLM predators vs. rules-only prey" is a first-class controlled configuration. *(iii) No orchestrator, no approval, no decay*: there is no team- or species-level controller; each agent reasons only from what it sees; the cache is replaced whole each slow tick.

4 The liquid substrate

Agents are coupled through a dynamical connectome recomputed every tick,

$$W_{ij}(t) \;=\; K(\|q_i - q_j\|) \cdot \Gamma(s_i, s_j)\,, \tag{2}$$

a geometric kernel K over spatial state q_i gated by a state function Γ over internal state s_i[1]. The connectome is engine-internal and never wrapped as a contract; each agent receives only its own row's projection through an MCP-shaped inbound membrane, never the full matrix. Everything that *crosses* an agent boundary (`move`, `pick_up`, `kick_ball`, …) is an MCP-shaped tool with a shared schema and two interchangeable backends (an in-process fast path; a networked interop path), so a cognitive core is bit-identical whether it runs in-process at 60 Hz or as a standalone server. World objects are first-class and carry their own dynamics, so agents practise *niche construction* — they reshape the world their successors inherit. The platform ships three configurable worlds on one engine: a Simple Demo (L1, physics + connectome only — the zero-cognition control), Predator–Prey (L3), and Soccer (L3, a possession game). Each is launched and fully parameterised from one dialog, with the objective-function stance and the per-collective LLM toggle as the primary knobs — because those are the independent variables of the science.

5 Telehomeostasis as a compressed per-agent objective

Telehomeostasis — operational definition

A collective is *telehomeostatic* when each of its members carries an objective whose argument is the persistence/thriving of the *kind* (lineage, team) rather than the individual, and acts on it using only locally available information. It is the compression of a collective persistence objective into a per-agent rule[2] — not a directive issued by a controller.

The operational commitment is strong and deliberate: **there is no collective.** The LLM is bound to one agent and sees only what that agent sees, as in nature. Telehomeostasis is instantiated through three local ingredients: (a) a *persona* that defines success collectively ("what matters is the team scoring / my kind persisting"); (b) an *objective-function stack* whose top priorities serve the kind (e.g. a sacrificial `protect_kin` draw-fire Value that replaces individual `flee`; a `team_play` Value that prefers a forward pass to a personal shot); and (c) a *shared observable* — the scoreboard — written into each agent's perception so the slow loop can reason "we are losing, push." The score is observation only: the fast objective-function/valence path never reads it, and the LLM-synthesised valence remains a researcher read-out. Coordinated behaviour, if it appears, is *emergent from local reasoning toward a shared goal*, which is exactly the regime in which the meta-agent question becomes meaningful.

6 Experiments

Both experiments hold everything bit-identical across arms except the objective stance, run rules-only (no LLM) for reproducibility, and sweep eight seeds.

6.1 EXP-001 — teamwork vs. selfishness in 11-v-11 soccer

A telehomeostatic team (press as a unit, hold pitch shape, pass to forward outlets) plays an otherwise identical selfish team (chase the ball, dribble and shoot personally). The headline is *not* the final score but the **substrate-dependence of its sign**. Across five successive substrate revisions the measured effect was, in order: a false 2:1 "win" (a blind-punt artifact); a wash under realistic passing; hidden entirely under a closed goal; an absurd 215:0 runaway (a no-formation-reset artifact); and finally, on a substrate with a real kickoff (centre spot, formation reset, possession to the conceding side) that can no longer flatter either side, an honest and robust result: **telehomeostatic beats selfish 7/8 seeds, 23–2 aggregate (∼11×)**, with selfish–vs–selfish collapsing to near-null noise and a mirror-symmetry control (the advantage follows the *stance*, not a side) confirming it is not a geometric artifact. The methodological claim is first-class: *a collective-objective effect must survive a substrate that cannot flatter it before it earns belief.*

6.2 EXP-002 — sacrificial prey and species persistence

Here the result is a **refutation**, sharpened by the per-collective control. Replacing selfish prey flight with a sacrificial draw-fire objective does *not* buy species persistence in the low-predation regime — it costs it. With predators held selfish and only the prey stance varied (the clean per-collective test), telehomeostatic prey persist at 0.57× the selfish baseline and lose all 8/8 seeds; isolating the prey stance *strengthens* the refutation relative to the confounded both-species design (0.64×). The interpretation is the KT-honest one: the value of an altruistic objective is not intrinsic; it is contingent on whether it is the right compression for the environment's *binding constraint*. Where predation is not the limiting factor, sacrifice is all cost.

Experiment	Result	Status
EXP-001 soccer (real-kickoff substrate)	tele beats selfish 7/8, 23–2 (∼11×)	confirmed, substrate-gated
EXP-002 predator–prey (prey-only IV)	tele prey 0.57× selfish, 0/8	hypothesis refuted

Table 1: The two pre-registered experiments. Both are reported with their full substrate-revision history; the refutation and the substrate-dependence are treated as results, not failures.

A configuration-matrix sweep corroborates internal consistency: in soccer, the telehomeostatic advantage is side-independent (it wins as either team and is null in selfish–vs–selfish); in predator–prey, the prey stance dominates the outcome independently of the predator stance. An LLM-on validation against a local model confirmed the slow loop is non-blocking and that an agent, given the shared scoreboard, reasons explicitly about "the team's need to score" and self-modulates its own objective (`team_play`×2) and planner parameters — purely per-agent, with the rules-only opponent untouched.

7 The emergence of meta-agents

The deeper goal is meta-agency. In KT a system is an algorithmic agent when it carries a compressive model, a non-trivial scalar objective, and a planner that selects actions to maximise it[3,4]. A *meta-agent* is a collective that satisfies this same definition at its own scale — a coherent higher-order agent whose "body" is its members and whose actions are their coordinated behaviour.

Meta-agent emergence conjecture

A collective behaves as a KT meta-agent when its members each (i) internalise a *shared* objective (a compressed collective persistence target), (ii) observe a *shared* state signal, and (iii) reason locally toward that objective — with *no* central orchestrator. Meta-agency is then an emergent property to be measured, not an architecture to be imposed.

This is why Agent Souper forbids a collective orchestrator on principle. A telehomeostatic soccer team is a candidate meta-agent: a shared objective (score), a shared observable (the scoreboard each player perceives), purely local per-agent reasoning, and — empirically, in EXP-001 — coordinated behaviour that an otherwise-identical non-telehomeostatic collective cannot produce. We are careful about what this shows. EXP-001 is suggestive evidence that the *right* compressed collective objective yields collective competence; EXP-002 is the essential counterweight showing that the *wrong* one is strictly worse than selfishness. Neither yet demonstrates that the team *itself* carries a compressive model of its own situation in the formal sense — that is the open measurement programme: define and estimate, on the running soup, the collective's model compression and the algorithmic mutual information between its coarse-grained state and its own future, and ask when those cross the agenthood threshold[5,6]. The per-agent LLM that internalises a collective goal from a local observation is our proposed *route* to that emergence, not a proof of it.

8 Limitations and roadmap

Agent Souper is a Beta. Known limits, stated plainly: L2 (fused neural policy) is specified but unimplemented; the LLM's slow-loop self-tuning exposes the objective-function and planner channels but not yet the forward-model (simulator) parameters; in *compressed headless* sweeps the slow loop emits only a handful of times (the dual-rate back-pressure that keeps the engine real-time also rarefies LLM influence off real-time), so LLM-outcome studies need real-time pacing or much longer runs; and the meta-agent claim is, as stated, a measurement programme not yet a theorem. The roadmap follows from these: a collective-agenthood metric computed on the live soup; a forward-model self-tuning channel; networked-MCP lift of an L3 agent to a standalone server; and a richer object/field world for niche-construction studies.

9 Conclusion

Agent Souper makes the Agent Soups formalism executable and the telehomeostasis and meta-agent questions empirical. Its central design choices — objective function always external, a deterministic fast loop the LLM may only *re-weight* (not rewrite, not bypass), no collective orchestrator, and honestly substrate-tested experiments — are not incidental engineering; they are what make the science interpretable. The first results are already instructive: a collective objective can decisively beat selfishness (soccer) *or* be strictly worse than it (predator–prey), and which one occurs is contingent on the environment and, dangerously, on the substrate. That contingency — and the discipline of refusing to believe an effect until the substrate can no longer flatter it — is the finding we most want to carry forward as the platform grows toward measuring meta-agency directly.

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